

DYNAMIC PROGRAMMING · CATEGORY THEORY

The free category of a dynamic program

Dynamic programming as functors out of a free category

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From first-order equations to a higher-order operator

Economists write a dynamic program as first-order equations between named quantities, $a = \sigma(m)$ and $m' = Ra + \xi'$. What a solver computes is higher order: the Bellman operator $\mathbb{T}$ on a space of value functions. A first-order syntax tree has no node for $\mathbb{T}$ and no record of the variables that $\max$ and $\mathbb{E}$ bind.

Proposal (Shanker). The operator is not a term. It is the image of an arrow of a free category under a functor. Syntax is a quiver over a signature; semantics is a functor out of its free category; the devices economists use informally, stages, post-decision states and Sargent–Stachurski's factored programs, are natural transformations, restriction along inclusions of ordinals, and an image factorization; and the Yoneda lemma says which operations on value functions are the natural ones.

Everything in *Category Theory in Context*, chapters 1–6, is assumed. The economics is defined once, on the next slide. Result numbers follow the source decks and their companion note.

A sequential decision problem

- State spaces X are measurable. A **stochastic kernel** $K: X \rightsquigarrow X'$ assigns to each x a probability measure $K(x, -)$ on X' . **Value functions** are measurable $v: X \rightarrow \bar{\mathbb{R}}$, forming L^X . The **backward operator** of K is $(K^*v)(x) := \int K(x, dx') v(x')$, where the integral exists; $(K'K)^* = K^*K'^*$.
- A **policy** σ selects a feasible action at each state. Its **policy operator** is $T_\sigma v = r_\sigma + \beta K_\sigma^* v$, with reward r_σ and discount $\beta \in (0, 1)$. The **Bellman operator** is $\mathbb{T}v = \sup_\sigma T_\sigma v$; the value function is its fixed point.

Example A (buffer stock), used throughout. Cash $m > 0$; savings $a = \sigma(m) \in [0, m)$; income $\xi' \sim \nu$, independent across periods; next cash $m' = Ra + \xi'$; reward $u(m - \sigma(m))$. Then

$$(\mathbb{T}v)(m) = \sup_{0 < c \leq m} \left\{ u(c) + \beta \mathbb{E} v(R(m - c) + \xi') \right\}.$$

A dynamic program is a quiver over a signature

Definition (signature). A quiver $\mathcal{S}$ whose vertices are **types** and edges **symbols** $\mathbf{s} : \mathbf{X} \rightarrow \mathbf{X}'$, with a set of **decision symbols**, each carrying a nonempty set $\Sigma_{\mathbf{s}}$ of **policies**. The quiver $\mathcal{S}_{\Sigma}$ has the same vertices, an edge $(\mathbf{s}, \sigma)$ for each policy of each decision symbol and one edge $(\mathbf{s}, *)$ for every other symbol; $p : \mathcal{S}_{\Sigma} \rightarrow \mathcal{S}$ forgets the policy.

Definition (declaration). A quiver Q of **fields** with a quiver map $\tau : Q \rightarrow \mathcal{S}$, the **typing**: an object of $\text{Quiver}/\mathcal{S}$. Its **policy quiver** is the pullback $Q_{\Sigma} := Q \times_{\mathcal{S}} \mathcal{S}_{\Sigma}$: each decision edge e becomes the parallel family $(e, \sigma)_{\sigma \in \Sigma_{\tau e}}$.

Example A. Types $\mathbf{Xm}, \mathbf{Xa}$; the decision symbol $\mathbf{save} : \mathbf{Xm} \rightarrow \mathbf{Xa}$ with the policies as $\Sigma_{\mathbf{save}}$, and $\mathbf{income} : \mathbf{Xa} \rightarrow \mathbf{Xm}$. Over N periods, Q is the chain $m_0 \rightarrow a_0 \rightarrow m_1 \rightarrow \cdots \rightarrow a_{N-1}$, typed alternately.

$$m_0 \xrightarrow{d_0} a_0 \xrightarrow{\ell_0} m_1 \xrightarrow{d_1} a_1 \xrightarrow{\ell_1} \cdots \longrightarrow a_{N-1}$$

Models are functors out of the free category

$F \dashv U : \mathbf{Cat} \rightleftarrows \mathbf{Quiver}$ (Riehl, Example 4.1.13). The **terms** of the signature are the arrows of $F\mathcal{S}_\Sigma$.

Definition (algebra, model). An **algebra** of the signature in $\mathbf{C}$ is a functor $F\mathcal{S}_\Sigma \rightarrow \mathbf{C}$. A **model** of a declaration in $\mathbf{C}$ is a functor $D : FQ_\Sigma \rightarrow \mathbf{C}$, equivalently a quiver map $Q_\Sigma \rightarrow UC$: a carrier $D(j)$ for each field and an arrow $D(e, \sigma)$ for each edge, with no equations imposed. An algebra A induces the model $A \circ F(\text{pr}_2)$.

Models form the functor category $\mathbf{C}^{FQ_\Sigma}$, and a morphism of models is a natural transformation. (The decks' Proposition 1 is this adjunction; for large $\mathbf{C}$ read it as the explicit description of functors out of a free category.)

Remark. In Lawvere's terms this is functorial semantics with no equations and no products: a model is a representation of the quiver Q_Σ in $\mathbf{C}$. Arities, hence products, enter only with equations that read several fields (slide 21).

The forward model takes values in Markov kernels

$\text{Stoch} := \text{Meas}_{\text{Prob}}$, the Kleisli category of the Giry monad (Riehl, Example 5.2.11(iv)), a Markov category in Fritz's sense. Arrows $K : X \rightsquigarrow X'$ are kernels; δ_f is the deterministic kernel of a measurable map f .

Definition (forward model). $D_f : FQ_\Sigma \rightarrow \text{Stoch}$ sends each field j to a measurable **state space** X_j and each edge $(e, \sigma) : j \rightarrow i$ to a kernel $X_j \rightsquigarrow X_i$, the law of the target field given the source field under that policy. In Example A the decision edge goes to the deterministic kernel δ_σ of the savings policy; in the decks' Example E the portfolio decision edge goes to a kernel that also integrates the returns.

	IN GENERAL	EXAMPLE A, ONE PERIOD $M \rightarrow A \rightarrow M'$
field j	its state space X_j	$m, m' \mapsto M = (0, \infty)$; $a \mapsto A = [0, \infty)$
edge $(e, \sigma) : j \rightarrow i$	a kernel $X_j \rightsquigarrow X_i$	$d_\sigma \mapsto \delta_\sigma$; $\ell \mapsto P$, $P(a, B) := \nu\{\xi : Ra + \xi \in B\}$

Laws. $\text{Prob}(X) = \text{Stoch}(*, X)$, so the push-forward $\mu \mapsto \mu K$ of a law along a path is postcomposition: the represented functor $\text{Stoch}(*, -)$ restricted along D_f .

The backward model is a presheaf of value functions

Definition (backward model). $D_b: (FQ_\Sigma)^{\text{op}} \rightarrow \text{Set}$ sends a field j to L^{X_j} and an edge $(e, \sigma): j \rightarrow i$ to an operator $L^{X_i} \rightarrow L^{X_j}$ against the edge; in the time-separable case, with K the edge's forward kernel, it is the affine map $v \mapsto r_{e,\sigma} + \beta_e K^* v$, with reward $r_{e,\sigma} = 0$ on policy-free edges and discount $\beta_e > 0$. Functoriality is $(K'K)^* = K^* K'^*$.

- The period $m_t \rightarrow a_t \rightarrow m_{t+1}$ of Example A goes to
 $D_b(d_\sigma) D_b(\ell) = T_\sigma$: **the policy operator is the image of a path.**
 Here $D_b(\ell) = P^*$ is the income expectation and $D_b(d_\sigma): w \mapsto u(m - \sigma(m)) + \beta w(\sigma(m))$.
- Top row: Q_Σ for one policy. Middle: D_f , along the edges. Bottom: D_b , against them. No vertical arrows join the rows, so the diagram asserts no equation.

$$m \xrightarrow{d_\sigma} a \xrightarrow{\ell} m'$$

$$M \xrightarrow{\delta_\sigma} A \xrightarrow{P} M$$

$$L^M \xleftarrow{D_b(d_\sigma)} L^A \xleftarrow{D_b(\ell)=P^*} L^M$$

The two models are not naturally isomorphic; they are paired

1. D_f and D_b differ in domain and in codomain, so no natural transformation between them is defined, let alone a natural isomorphism.
2. The pairing $\langle \mu, v \rangle := \int v \, d\mu$ satisfies $\langle \mu K, v \rangle = \langle \mu, K^* v \rangle$ for every kernel K , by Fubini, where both sides are finite. Hence $\mu \mapsto \langle \mu, - \rangle$ is natural, $\mathbf{Stoch}(*, -)D_f \Rightarrow \mathbf{Set}(D_b(-), \bar{\mathbb{R}})$, the analogue of evaluation into the double dual (Riehl, Example 1.4.4(i)); equivalently the pairings $\langle -, - \rangle_j$ on $\mathbf{Prob}(X_j) \times L^{X_j}$ form a **dinatural** transformation (Riehl, footnote 34 and Exercise 1.4.viii).
3. With rewards, $\langle \mu, D_b(e, \sigma)v \rangle = \langle \mu, r_{e, \sigma} \rangle + \beta_e \langle \mu K, v \rangle$ is affine in v and is not a dinaturality condition.
4. For finite state spaces K is a stochastic matrix and the identity reads $(\mu K)v = \mu(Kv)$.

Groupings are inclusions of ordinals

A **chain** is a declaration whose quiver is $0 \rightarrow 1 \rightarrow \dots \rightarrow N$. With one policy fixed at each decision edge, its free category is the ordinal $[N] = \{0 < \dots < N\}$ (Riehl's $\mathbb{N}+1$, Example 4.1.14). Example A over N periods is a chain with $2N - 1$ edges, positions $m_0, a_0, m_1, \dots, a_{N-1}$.

Definition (grouping, grouped model, stage structure). A **grouping** $j_0 < \dots < j_n$ is a monomorphism $\pi: [n] \hookrightarrow [N]$, a composite of cofaces d^i . The **grouped model** of D is the restriction $\pi^* D = D\pi$. A **stage structure** is a grouping with $\pi(0) = 0$ and $\pi(n) = N$.

Grouping changes the category on which the model is defined, so it is not a natural transformation, which compares two functors on one category. Natural transformations between grouped models appear on the next slides.

A stage contains exactly one decision

A **run** is a path of policy-free edges, the empty path included. A **stage** is an edge of the grouped quiver whose path contains exactly one decision edge: from its **arrival field** through a run to the decision edge, then through a run to its **continuation field**. A **staging** is a stage structure all of whose edges are stages. A boundary may move across a run, never across a decision edge.

Lemma (number of stagings). With $n_d \geq 1$ decision edges and n_k edges in the run between the k -th and $(k + 1)$ -st, a chain has $\prod_{k=1}^{n_d-1} (n_k + 1)$ stagings, which is 1 when $n_d = 1$.

Under D_b a stage goes to $T_\sigma = U \cdot D_b(e, \sigma) \cdot F$: the **continuation map** F and the **arrival map** U are the operators of the two runs and do not depend on σ . This is the **stage form** of $\{T_\sigma\}$; in the Bellman project it is the arrival–decision–continuation structure of a stage.

$$\text{stage path in } Q_\Sigma : \quad j_0 \xrightarrow{\text{run}} i_0 \xrightarrow{(e, \sigma)} i_1 \xrightarrow{\text{run}} j_1$$

$$\text{backward model:} \quad L^{X_{j_0}} \xleftarrow{U := D_b(j_0 \rightarrow i_0)} L^{X_{i_0}} \xleftarrow{r_{e, \sigma} + \beta_e D_f(e, \sigma)^*} L^{X_{i_1}} \xleftarrow{F := D_b(i_1 \rightarrow j_1)} L^{X_{j_1}}$$

Sargent–Stachurski's factored programs are naturality squares

A **factored dynamic program** $(V, F, \hat{V}, \{G_\sigma\}_{\sigma \in \Sigma})$ defines $T_\sigma := G_\sigma F$ and $\hat{T}_\sigma := F G_\sigma$, so $FT_\sigma = \hat{T}_\sigma F$. For each $\sigma, n \mapsto T_\sigma^n$ and $n \mapsto \hat{T}_\sigma^n$ are functors $\mathbf{BN} \rightarrow \mathbf{Set}$ and F is a natural transformation between them: the square at n follows from the square at 1 by pasting (Riehl, Lemma 1.6.11).

Proposition 5. On a chain of periods oriented as the operators act, with edges $(t, \sigma): t+1 \rightarrow t$, $D(t) = V_t$ and $D(t, \sigma) = T_{t,\sigma}$, suppose $T_{t,\sigma} = G_{t,\sigma} F_{t+1}$ with F_{t+1} independent of σ ; put $F_0 := \text{id}$. Then the F_t are the components of a natural transformation $F: D \Rightarrow \hat{D}$ to the **reduced model** $\hat{D}(t) := \hat{V}_t$, $\hat{D}(t, \sigma) := F_t G_{t,\sigma}$, and for every policy sequence

$$F_0 T_{0,\sigma_0} \cdots T_{N-1,\sigma_{N-1}} = \hat{D}(0, \sigma_0) \cdots \hat{D}(N-1, \sigma_{N-1}) F_N.$$

Since $F_0 = \text{id}$, the reduced model returns the same date-0 value function for every policy sequence. In Example A, F is the income expectation and $\hat{V}$ consists of value functions of savings; Sargent and Stachurski call the two systems strongly semiconjugate under F, G_σ .

Moving a boundary is whiskering a 2-cell of posets

Proposition 4. Let D be a model of a chain with one policy fixed at each decision edge, and $\pi, \pi' : [n] \rightarrow [N]$ groupings with $\pi(i) \leq \pi'(i)$ for every i . The arrows $D(\pi i \rightarrow \pi' i)$ are the components of a natural transformation $\pi^* D \Rightarrow \pi'^* D$. If $\pi(i) > \pi'(i)$ for some i , the family is not defined.

Proof. $[N]$ is a poset, so $\pi \leq \pi'$ is the unique natural transformation $\pi \Rightarrow \pi'$; whisker it with D (Riehl, Remark 1.7.6). $\square$

- For the contravariant D_b the inequality is read in the reversed order: the components run from the grouping whose boundaries are later in Q to the one whose boundaries are earlier.
- **Example A.** Moving every interior boundary from cash m_{t+1} to savings a_t gives $D_b(\text{cash staging}) \Rightarrow D_b(\text{savings staging})$, with components the income expectations $D_b(\ell_t)$.

Transforming some fields: descent to a quotient

Proposition 6 (partial transformations). Let $D: FQ_\Sigma \rightarrow \mathbf{Set}$ be a model (for D_b , reverse every edge), J a set of fields, and $\varphi_j: D(j) \twoheadrightarrow B_j$ onto for $j \in J$; put $B_j := D(j)$ and $\varphi_j := \text{id}$ for $j \notin J$. There exist a model D' with $D'(j) = B_j$ and a natural transformation $\varphi: D \Rightarrow D'$ with these components if and only if, for every edge $(e, \sigma): j \rightarrow i$ with $j \in J$ and all $v, v' \in D(j)$,

$$\varphi_j(v) = \varphi_j(v') \quad \text{implies} \quad \varphi_i(D(e, \sigma) v) = \varphi_i(D(e, \sigma) v').$$

Then D' is unique, with $D'(e, \sigma)(\varphi_j v) = \varphi_i(D(e, \sigma) v)$. Such a family is **admissible**.

In other words, $\varphi_i D(e, \sigma)$ coequalizes the kernel pair of φ_j , so $D(e, \sigma)$ descends to the quotient B_j . The condition is local: one field and one edge at a time.

The coarsest cut is an image factorization

For maps $T_\sigma: V \rightarrow W, \sigma \in \Sigma$, a **cut** is $(\hat{V}, F, G_\sigma)$ with $T_\sigma = G_\sigma F$ and F independent of σ ; it is **coarser** than $(\hat{V}', F', G'_\sigma)$ when $F = \omega F'$ for some $\omega: F'(V) \rightarrow \hat{V}$. Write $v \sim v'$ when $T_\sigma v = T_\sigma v'$ for every σ .

Theorem 7 (the coarsest cut). Let $\bar{V} := V/\sim, q: V \twoheadrightarrow \bar{V}$ the quotient map and $\bar{G}_\sigma[v] := T_\sigma v$. Then $(\bar{V}, q, \bar{G}_\sigma)$ is a cut, and for every cut $(\hat{V}, F, G_\sigma)$ there is exactly one $\omega_F: F(V) \rightarrow \bar{V}$ with $q = \omega_F F$; it is onto, and $\bar{G}_\sigma \omega_F = G_\sigma$ on $F(V)$. Hence a coarsest cut exists and is unique up to a unique bijection.

Proof. $(\bar{V}, q)$ is the image factorization of $\langle T_\sigma \rangle_\sigma: V \rightarrow W^\Sigma$ (Riehl, Example 3.6.11); a cut is a factorization of this map through F , and the image is the smallest quotient of V through which it factors. $\square$

Example A. $v \sim v'$ iff $\mathbb{E} v(Ra + \xi') = \mathbb{E} v'(Ra + \xi')$ for every a : q is the income expectation and $\bar{V}$ its image, the economists' value function of the **post-decision state**. When ν has a density, q is not injective, so this is a genuine quotient and not a natural isomorphism; its use is exactly that it discards what no policy operator can distinguish.

3 STAGES · THE CANONICAL STAGE FORM

Every stage form factors through the canonical one

1. Let $W_\Sigma := \bigcup_\sigma T_\sigma(V) \subseteq W$, with inclusion U_Σ , and read $\bar{G}_\sigma$ as a map $\bar{V} \rightarrow W_\Sigma$. A **stage form** is a factorization $T_\sigma = UG_\sigma F$ with $F: V \rightarrow \hat{V}$ and $U: \tilde{V} \rightarrow W$ independent of σ . The top row $T_\sigma = U_\Sigma \bar{G}_\sigma q$ is one.
2. Two equations link every stage form to it through the single map $\omega_F: q = \omega_F F$ on V , by Theorem 7 for the cut $(\hat{V}, F, UG_\sigma)$; and $U_\Sigma \bar{G}_\sigma \omega_F = UG_\sigma$ on $F(V)$, since both sides send Fv to $T_\sigma v$.
3. Dually, U_Σ is the image of the copairing $[T_\sigma]_\sigma: \Sigma \times V \rightarrow W$: for every factorization $T_\sigma = U\bar{G}_\sigma$ with U monic and independent of σ there is exactly one $\omega_U: W_\Sigma \rightarrow \tilde{V}$ with $U\omega_U = U_\Sigma$ (Proposition 8 of the companion note).
4. The family $\{T_\sigma\}$ determines neither the F nor the U of a declaration: q is the coarsest F and U_Σ the finest U . Each factor of a stage is order-preserving for the pointwise order, so every policy operator of a stage is.

$$\begin{array}{ccccccc}
 V & \xrightarrow{q} & \bar{V} & \xrightarrow{\bar{G}_\sigma} & W_\Sigma & \xrightarrow{U_\Sigma} & W \\
 \parallel & & \uparrow \omega_F & & & & \parallel \\
 V & \xrightarrow{F} & F(V) & \xrightarrow{G_\sigma} & \tilde{V} & \xrightarrow{U} & W
 \end{array}$$

From a backward factorization to a forward one

1. **Forward to backward is immediate.** A factorization $K = K_2 K_1$ of a kernel gives $K^* = K_1^* K_2^*$, and a grouping groups both models alike, as $\pi^* D_f$ and $(\pi^{\text{op}})^* D_b$.
2. **Backward to forward is not.** In a cut $T_\sigma = G_\sigma F$ the map F may be any map of sets, for instance $Fv := v^3$ pointwise. On bounded functions, a map $L^{X'} \rightarrow L^X$ is K^* for a kernel K iff it is linear, positive, unital and continuous under bounded monotone limits; then $K(x, B) = (T\mathbf{1}_B)(x)$.
3. **For a stage,** on functions with finite integrals, $T_\sigma v - T_\sigma v' = \beta_\sigma K_\sigma^*(v - v')$, so $v \sim v'$ iff $K_\sigma^* v = K_\sigma^* v'$ for every σ , and q is the backward operator of the kernel $J((\sigma, x), -) := K_\sigma(x, -)$ on pairs (policy, arrival state), corestricted to its image. In Example A the reduced model is itself a backward model, on functions of savings.
4. **Isomorphisms.** A component is a bijection when it is the backward operator of an invertible deterministic map, as in the decks' perfect-foresight application; a bijective K^* need not be deterministic, since the stochastic matrix with rows $(0.9, 0.1)$ and $(0.1, 0.9)$ is invertible.

Policies are sections; inserting one is precomposition

Let $E := \{(s, b) : s \in S, b \in \mathcal{D}(s)\}$ with $p: E \rightarrow S, p(s, b) := s$. Every section of p is $\iota_\sigma(s) = (s, \sigma(s))$ for a feasible policy σ : sections are exactly the feasible policies. In the reward-free backward model the decision operator is $\iota_\sigma^*: \mathbf{Set}(E, Z) \rightarrow \mathbf{Set}(S, Z), w \mapsto w\iota_\sigma$.

Proposition 10. A family $\Phi_Z: \mathbf{Set}(E, Z) \rightarrow \mathbf{Set}(S, Z)$ natural in Z is ι^* for exactly one $\iota: S \rightarrow E$, namely $\iota = \Phi_E(\text{id}_E)$; and ι is a section of p if and only if $\Phi_S(p) = \text{id}_S$. (Riehl, Corollary 2.2.8 with $c = E$ and $c' = S$.)

Expectation is **not** natural in the value set: for $\lambda(z) := z^2$, averaging 0 and 2 then squaring gives 1; squaring then averaging gives 2. It is natural for affine λ , since a probability integrates to one. So T_σ , which adds a reward and takes an expectation, is not of the form ι^* .

Rules: the stage form is the form of every natural operation

For $\mathbf{C}$ locally small and objects c_0, c_1, c_2, c_3 , a **rule** is a family of maps $\alpha_P: P(c_1, c_2) \rightarrow P(c_0, c_3)$, one for each bifunctor $P: \mathbf{C}^{\text{op}} \times \mathbf{C} \rightarrow \mathbf{Set}$, natural in $P: \kappa_{(c_0, c_3)} \alpha_P = \alpha_{P'} \kappa_{(c_1, c_2)}$ for every $\kappa: P \Rightarrow P'$.

Proposition 11 (companion note). Every rule is $\alpha_P = P(F, U)$ for exactly one pair of arrows $F: c_0 \rightarrow c_1$ and $U: c_2 \rightarrow c_3$.

Proof. The Yoneda argument for the representable $(\mathbf{C}^{\text{op}} \times \mathbf{C})((c_1, c_2), -) = \mathbf{C}(-, c_1) \times \mathbf{C}(c_2, -)$ and its element $(\text{id}_{c_1}, \text{id}_{c_2})$: a rule is determined by $\alpha(\text{id}_{c_1}, \text{id}_{c_2}) = (F, U)$. The proof of Theorem 2.2.4 uses only that the transformations out of a representable form a set, which the lemma itself supplies, so $\mathbf{C} = \mathbf{Set}$ is admitted although $\mathbf{Set}^{\mathbf{C}^{\text{op}} \times \mathbf{C}}$ is not locally small. $\square$

On the hom bifunctor the rule sends $G: c_1 \rightarrow c_2$ to UGF . So a factorization $T_\sigma = UG_\sigma F$ with the same F and U for every σ , the stage form, is the only way of varying with the policy that is natural in this sense.

What the lemma does, and does not, do here

It does.

- Propositions 10 and 11 characterize the operations inside a stage that are natural in the value set and in the bifunctor: precomposition with a section, and conjugation by fixed maps.
- The initiality of $(\mathbb{N}, s, 0)$ (Riehl, Example 2.1.1) and of (FQ, η) (Proposition 1) are initial-object statements; Proposition 2.4.8 adds that 0 is a universal element. For (FQ, η) the lemma is not applied: interpretations in **Set** or **Stoch** live in a category of categories that is not locally small.

It does not.

- Choose a factorization: $F = U = \text{id}$ with $G_\sigma = T_\sigma$ is a stage form.
- Produce the quotient q , which comes from images, not from the lemma; by Proposition 6, q is the coarsest admissible component at one field.
- Make precise the resemblance between knowing a declaration through its models and knowing an object by the arrows out of it (Riehl, Proposition 2.3.1); the decks leave it informal.

Where this sits

- **Functorial semantics.** Models as functors (Lawvere 1963; Riehl, §5.5), here with no equations and no products: representations of a quiver. The syntax-to-semantics step follows Jacobs's classifying categories.
- **Categorical probability.** Stoch as the Kleisli category of Giry's monad; Markov categories (Fritz 2020).
- **Categorical systems theory.** Open dynamical systems as lenses and double categories: Myers, *Categorical Systems Theory*, Spivak and coauthors (wiring-diagram operads, 2015; sheaves, 2019; **Poly**, 2020; Niu and Spivak, 2025); AlgebraicDynamics (Libkind, Baas, Patterson and Fairbanks, 2021). Those frameworks compose systems along interfaces. Here the composite is fixed by the declaration, and the question is which functors out of it, and which transformations between them, carry the economics.
- **Open games and optics.** Ghani, Hedges, Winschel and Zahn (2018); Hedges and Rodríguez Sakamoto, value iteration as optic composition (2022) and reinforcement learning in categorical cybernetics (2025), where Bellman updates are parametrised optics and a representable functor does the work. Bakirtzis, Savvas and Topcu (2025): a category of MDPs with pushouts for task composition.
- **Decisions by Kan extension.** Mahadevan, *Universal decisions with Kan extensions*.

Shanker's own assessment: "not novel category theory; the applied mathematics is." The claim is a syntax for dynamic programs whose semantics is a functor, and the recognition of the economists' devices as the constructions above.

Not yet done

1. **Several inputs.** An equation that reads several fields is not an edge of a quiver. Symbols need arities, interpreted by products of state spaces; the product of measurable spaces is not a categorical product in Stoch , because a kernel into $X \times Y$ is not determined by its two marginals.
2. **Branching.** A discrete choice among continuations reads several fields, so it needs item 1. Shanker: "branching is where the role of initial diagrams becomes really powerful."
3. **Dimension.** In Example A the quotient q moves from functions of cash to functions of savings, both of one variable. Which declarations have coarsest cuts on functions of fewer variables than the state is not examined.
4. **Choice of staging.** Among the $\prod_k (n_k + 1)$ stagings one may seek those with the fewest distinct stage terms, or those whose coarsest cuts are smallest; neither criterion is examined. A **rotation** of a closed term (a term with equal source and target type, cyclically shifted, so that the period starts at another field) gives operators related by a natural transformation under every model (Proposition 4); an exchange of two symbols does not.

Questions for a category theorist

1. Is "a quiver over a signature quiver, with models the functors out of $F(Q \times_{\mathcal{S}} \mathcal{S}_{\Sigma})$ " the right syntax, or should it be a sketch or a finite-product theory from the start, to admit arities and branching?
2. The backward model is a presheaf on FQ_{Σ} built from $\mathbf{Meas}(-, \bar{\mathbb{R}})$ through the Giry monad. Is there a clean statement, as a Kleisli extension against a barycentre algebra, a profunctor, or an extranatural transformation, that also carries the rewards?
3. The Bellman operator $\mathbb{T} = \bigvee_{\sigma} T_{\sigma}$ is a join over a parallel family of arrows. Is the right setting **Pos**- or **Sup**-enriched models, with optimality a colimit-like operation on hom-posets?
4. Stagings are restrictions along cofaces. Do $\mathbf{Lan}_{\pi}$ and $\mathbf{Ran}_{\pi}$ have economic meaning, as canonical refinements or coarsenings of a stage structure?
5. The coarsest cut is an image in **Set**. In which categories of value functions (bounded, measurable, L^p) does the (epi, mono) factorization still give the economists' reduction?
6. For branching: tree-shaped quivers, colimits, or initial algebras of polynomial functors as in the systems-theory literature?
7. Is the category of elements $\int D_b$, whose objects are a field with a value function and whose morphisms are the policy paths realizing it, a useful object? Backward induction walks in it.

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